Network Science

Lecture 06: Structural Balance

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From Unsigned to Signed Graphs

So far, an edge meant "connected." Starting now, every edge also carries a **sign**: positive (+) for alliance, friendship, or trust; negative (−) for rivalry, hostility, or distrust.

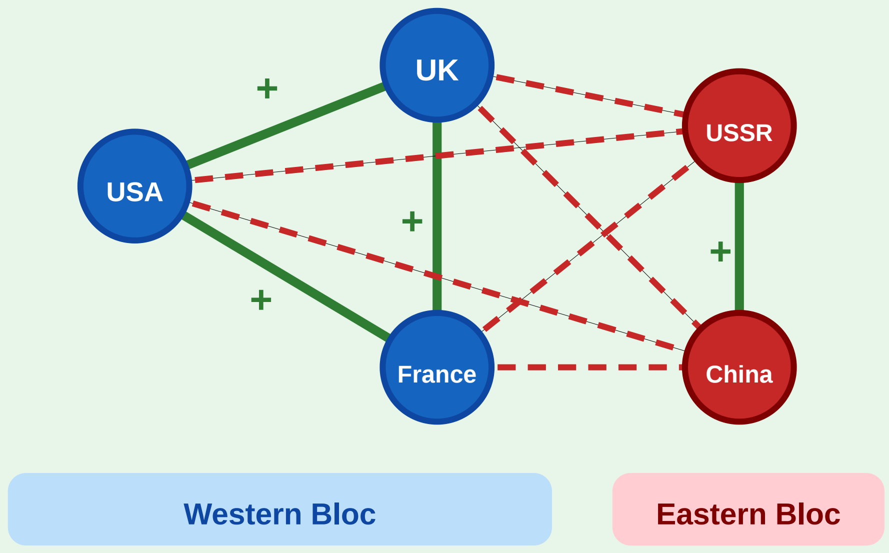
This one addition — a single bit per edge — turns out to constrain global structure powerfully.

A Motivating Scenario

Cold War alliances as a signed graph

green solid = alliance (+) · red dashed = rivalry (−)

Pre-1960 — two balanced blocs

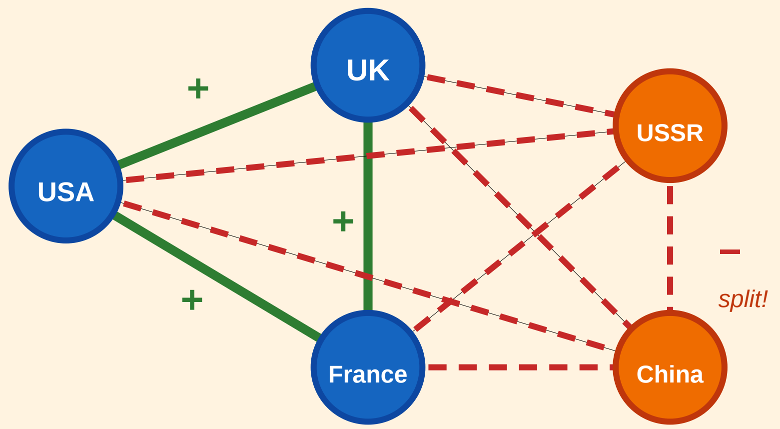


All 10 triangles are balanced

→ Balance Theorem: exactly 2 camps

✓ strong structural balance

Post-1969 — Sino-Soviet split



△ USA–USSR–China: all negative (−−−)

→ violates strong balance

→ but satisfies weak balance (≥3 camps)

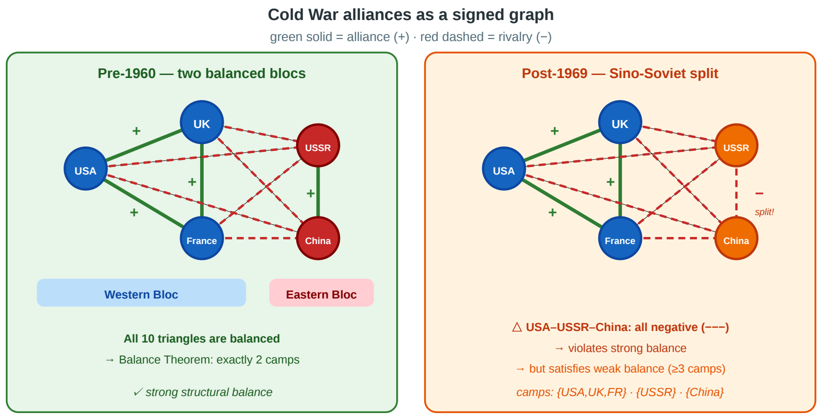
camps: {USA,UK,FR} · {USSR} · {China}

Five nations during the Cold War. **Left:** pre-1960, all alliances and rivalries form two clean blocs — every triangle is balanced. **Right:** after the 1969 Sino-Soviet split, USSR–China flips to −, creating all-negative triangles. The two-bloc structure breaks.

Four Questions from One Picture

1. **Local stability.** Which signed triangles feel psychologically "stable" — and which feel like they should change?
2. **Global structure.** If every triangle is stable, what does that force the *whole* graph to look like?
3. **Relaxation.** When a three-way rivalry (all −) appears, does the two-bloc theorem still hold — or do we need more camps?
4. **Measurement.** In a real signed network with millions of edges, how close to balanced is it — and how would we know?

Every concept in today's lecture answers one of these.



A New Kind of Gap

L03–L04 had a **computational** gap: MaxSTC and modularity maximization are NP-hard. L05 had a **causal** gap: cross-sectional data cannot identify selection vs. socialization.

L06 has a **structural** gap: the Balance Theorem requires a *complete* signed graph, but real networks are *sparse*. Most pairs have no observed edge, so the theorem's premise is never met exactly.

Lecture	Gap type	Ideal	Reality	Strategy
L03–L04	Computational	Exact optimization	NP-hard	Polynomial heuristics
L05	Causal	Mechanism identification	Snapshots insufficient	Longitudinal data
L06	Structural	Complete signed graph	Sparse, noisy data	Approximate balance measures

Takeaway

> Adding a sign to each edge creates a surprisingly powerful local constraint. The structural balance theorem turns triangle-level psychology into a global two-bloc prediction — but applying it to real data requires dealing with incompleteness and approximation.

Signed Graphs and Structural Balance

> **Guiding Question:** If every triangle in a signed graph is "psychologically stable," what does that forc
> the global structure to be?

Heider's Insight (1946)

Fritz Heider (1946), *Attitudes and Cognitive Organization*, Journal of Psychology 21, 107–112.

Heider proposed that people experience psychological tension ("cognitive dissonance") when their relationships form certain patterns — for example, two friends who both dislike the same person feel stable, but two friends who disagree about a third person feel pressure to change.

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The network-science formalization: encode the tension as a constraint on **signed triangles**.

Signed Graphs — Formal Definition

> A **signed graph** is a pair (G, σ) where $G = (V, E)$ is a graph and $\sigma : E \to +, -$ assigns a sign to every edge.
>
> - $\sigma(e) = +$: alliance, friendship, trust
> - $\sigma(e) = -$: rivalry, hostility, distrust

A **complete signed graph** has an edge (with some sign) between every pair of nodes.

The Four Signed Triangles

The four signed triangles: +++, ++−, +−−, −−−

A triangle is **balanced** if and only if it has an **even** number of negative edges (0 or 2). The $(+,+,-)$ triangle is unstable: two friends share an enemy, creating pressure for one friendship to break. The $(-,-,-)$ triangle is the ambiguous case — stable under weak balance but not strong.

The Structural Balance Theorem

> **Structural Balance Theorem (Cartwright & Harary, 1956).** A complete signed graph (G, σ) is balanced — i.e., every triangle has an even number of negative edges — if and only if the nodes can be partitioned into **at most two** groups such that:
>
> - every edge *within* a group is positive
> - every edge *between* groups is negative

In other words: **balance implies two camps**.

Proof Sketch

Direction 1 (camps → balance): If the graph splits into ≤2 camps with + inside and − between, then every triangle has 0 or 2 negative edges (check the three possible placements of three nodes across two groups). ✓

Direction 2 (balance → camps): Pick any node v, and put all its friends ($+$ neighbors) in group A with v, and all its enemies ($-$ neighbors) in group B. Now take any two nodes u, w in group A. They are both friends of v, so the triangle v–u–w has two positive edges (v–u and v–w). For the triangle to be balanced, u–w must be $+$. By symmetric reasoning, any two nodes in B are linked by $+$, and any pair $(a \in A, b \in B)$ by $-$. ✓

Back to the Cold War

In the pre-1960 panel: **all 10 triangles** in K_5 are balanced.

Triangle	Signs	Count of −	Balanced?
USA–UK–France	$+, +, +$	0	✓
USA–UK–USSR	$+, -, -$	2	✓
USA–UK–China	$+, -, -$	2	✓
USA–France–USSR	$+, -, -$	2	✓
USA–France–China	$+, -, -$	2	✓
USA–USSR–China	$-, -, +$	2	✓
UK–France–USSR	$+, -, -$	2	✓
UK–France–China	$+, -, -$	2	✓
UK–USSR–China	$-, -, +$	2	✓

Triangle	Signs	Count of −	Balanced?
France–USSR–China	$-, -, +$	2	✓

The theorem correctly predicts two camps: {USA, UK, France} and {USSR, China}.

Takeaway

> The Structural Balance Theorem is a rare example of a local constraint (triangle psychology) producing a clean global prediction (two camps). A single triangle rule, applied consistently across a complete signed graph, locks the entire network into a rigid two-bloc structure.

Quick Check

Consider four nations with the following signed edges: A–B: +, A–C: +, A–D: −, B–C: +, B–D: −, C–D: −.

1. List all four triangles and classify each as balanced or unbalanced.
2. Is the graph balanced? If so, what are the two camps?
3. If we flip the C–D edge from − to +, which triangle(s) become unbalanced?

Quick Check — Solution

1. Triangle classification:

Triangle	Signs	Balanced?
A–B–C	$+, +, +$	✓
A–B–D	$+, -, -$	✓
A–C–D	$+, -, -$	✓
B–C–D	$+, -, -$	✓

2. All balanced → . Camps: A, B, C vs D.

Flipping C–D to + makes:

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- A–C–D → $+, +, +$ ✓ (still balanced)

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B–C–D	$+, -, -$	✓

2. All balanced → **yes**. Camps: A, B, C vs D.

3. Flipping C–D to + makes:

- A–C–D → $+, +, +$ ✓ (still balanced)
- B–C–D → $+, +, -$ ✗ (unbalanced — B and C are friends, C and D are now friends, but B and D are enemies)

Weak Structural Balance

> **Guiding Question:** What happens to the two-camps theorem when three-way rivalries are allowed?

The Problem with All-Negative Triangles

In the post-1969 Cold War panel, the triangle USA–USSR–China has three negative edges.

Under strong balance, this is **unbalanced** — the theorem says the graph cannot split into two camps. But historically, the world *did* function with three separate power blocs: {USA, UK, France}, {USSR}, {China}.

The Problem with All-Negative Triangles

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The issue is psychological: a three-way rivalry may be *unstable* (each party would benefit from allying with someone), but it is not *contradictory* in the way that $(+, +, -)$ is. Two friends disagreeing about a third face direct tension; three mutual enemies merely face *opportunity*.

Weak Balance — Definition

Weak Structural Balance (Davis, 1967). A complete signed graph is **weakly balanced** if it contains no triangle with exactly **one** negative edge $(+, +, -)$. All-negative triangles $(-, -, -)$ are permitted.

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Weak Balance Theorem. A complete signed graph is weakly balanced if and only if the nodes can be partitioned into $k \geq 1$ groups such that:

- every edge *within* a group is positive
- every edge *between* groups is negative

(Strong balance is the special case $k \leq 2$.)

Back to the Cold War — Post-1969

After the Sino-Soviet split, the five-nation graph has:

- $(+,+,+)$ triangles: USA–UK–France ✓
- $(+,-,-)$ triangles: all Western-pair + one Eastern node (6 triangles) ✓
- $(-,-,-)$ triangles: USA–USSR–China, UK–USSR–China, France–USSR–China ✓ under weak balance

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No $(+,+,-)$ triangles exist → **weakly balanced**.

Partition into 3 camps: $\text{USA, UK, France}, \text{USSR}, \text{China}$.

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Takeaway

Weak balance replaces the two-bloc theorem with a k-coalition theorem. Permitting all-negative triangles allows multiple hostile camps — a better model for real-world multipolar structures like the post-1969 Cold War.

Quick Check

A social network of 6 people has the following signed edges (complete graph):

- Within A, B: all +
- Within C, D: all +
- Within E, F: all +
- Between any two groups: all −

1. Is this graph balanced (strong)? Why or why not?
2. Is it weakly balanced?
3. What is the minimum number of edge flips needed to make it strongly balanced?

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Quick Check — Solution

Triangles like A–C–E have three negative edges — forbidden under strong balance.

No $(+,+,-)$ triangle exists. Three camps: A, B, C, D, E, F.

Strong balance requires ≤2 camps. To merge {C,D} into {A,B}, flip the 4 cross-group edges (A–C, A–D, B–C, B–D) from − to +. Then the two camps are {A,B,C,D} and {E,F}.

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3. Minimum 4 flips. Strong balance requires ≤2 camps. To merge {C,D} into {A,B}, flip the 4 cross-group edges (A–C, A–D, B–C, B–D) from − to +. Then the two camps are {A,B,C,D} and {E,F}.

Empirical Test: Signed Networks in Social Media

> **Guiding Question:** In real signed networks with millions of edges, does balance actually hold?

The Study

Leskovec, Huttenlocher & Kleinberg (2010), *Predicting Positive and Negative Links in Online Social Networks*, Proc. 19th International Conference on World Wide Web (WWW), 641–650.

They analyzed three large signed networks:

Platform	Nodes	Signed edges	Type
Epinions	131 828	841 372	trust / distrust
Slashdot	82 144	549 202	friend / foe
Wikipedia	7 118	103 747	support / oppose (admin elections)

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What They Found

Triangle-level balance is remarkably strong:

Triangle type	Expected (random signs)	Observed (Epinions)
$+, +, +$	~12.5%	~47%
$+, +, -$	~37.5%	**~8%**
$+, -, -$	~37.5%	~37%
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The $(+, +, -)$ type — the one both strong and weak balance forbid — is **massively underrepresented** (~8% vs. 37.5% expected). The balanced types $(+, +, +)$ and $(+, -, -)$ are overrepresented. Balance is not perfect, but the skew is dramatic.

What the Study Could Not See

1. Sparse, not complete. The Balance Theorem assumes a complete signed graph. These networks are extremely sparse — most pairs have no edge. The theorem's prediction (two or k camps) does not directly apply; the study tests triangle statistics, not the global partition.

2. Directed signs. On Epinions, trust and distrust are *directed* — A can trust B without B trusting A. The balance framework assumes undirected signs. Directed balance theory exists (Leskovec et al. explore it) but is less clean.

3. Missing negative edges. People are less likely to *create* a negative edge than to simply *not interact*. The observed negative edges are a biased sample: they represent actively expressed hostility, not indifference. The "true" signed graph — if every pair had a relationship — might look different.

4. Temporal dynamics. The data is a snapshot. Balance theory predicts an *equilibrium*; the snapshot might capture a network mid-transition, with unbalanced triangles that are in the process of resolving.

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Takeaway

Leskovec et al. (2010) showed that balance holds at the triangle level in large online signed networks: the forbidden $(+, +, -)$ type is massively underrepresented. But real networks are sparse and directed, so the clean two-camp prediction of the Balance Theorem must be tested through approximate measures, not exact partition.

Incomplete and Approximate Balance

Guiding Question: How can we assess balance when the graph is incomplete or only approximately balanced?

From Triangles to Cycles

On a complete graph, checking triangles is sufficient for balance (the theorem guarantees it). On an *incomplete* graph, triangles alone are not enough — balance must be checked on **all cycles**.

> **Cycle criterion.** A signed graph (G, σ) is balanced if and only if every cycle in G has an **even** number of negative edges.
>
> Equivalently: no cycle has an odd number of negative edges.

The Frustration Index

In real data, perfect balance is rare. The natural question becomes: **how far** is the graph from balanced?

> **Frustration index** $F(G, \sigma)$: the minimum number of edges whose sign must be flipped to make the graph balanced.
>
> $$F(G, \sigma) = \min_{\sigma'} |e \in E : \sigma(e) \neq \sigma'(e)|$$
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> subject to (G, σ') being balanced.

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Normalized frustration: $f = F/|E|$. If $f \approx 0$, the graph is near-balanced; if $f \approx 0.5$, it is no more balanced than a randomly signed graph.

Computing Approximate Balance

Exact computation. Finding F exactly is NP-hard (equivalent to a minimum-weight graph cut on a related unsigned graph). On small graphs, integer programming or brute force works; on large graphs, we need heuristics.

Spectral approach. The **signed Laplacian** $L_\sigma = D - A_\sigma$ (where A_σ has entries $+1$ or -1) has the property that (G, σ) is balanced iff the smallest eigenvalue $\lambda_1(L_\sigma) = 0$. The magnitude of λ_1 quantifies how far the graph is from balanced.

Practical rule of thumb. For large networks, count the fraction of balanced triangles ($T_{\text{bal}}/T_{\text{total}}$) as a fast proxy. Leskovec et al.'s Epinions data had $T_{\text{bal}}/T_{\text{total}} \approx 0.92$.

Quick Check

A signed graph on 5 nodes has 7 edges. You compute that 8 out of 10 triangles are balanced and 2 are unbalanced (both of type $+, +, -$).

1. Is the graph balanced? Why or why not?
2. Can you conclude that the frustration index is exactly 2?
3. What additional information would you need to determine the exact frustration index?

Quick Check — Solution

Two unbalanced triangles violate the "every triangle must be balanced" condition.

2. No. Frustration index counts , not unbalanced triangles. The two bad triangles might share an edge — flipping that one shared edge could fix both. So $F \leq 2$, but F could be as low as 1.

— which nodes and edges form the unbalanced triangles, and whether they share edges. Determining F exactly requires solving a minimum-cut problem on the graph (NP-hard in general, but tractable on small instances).

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2. No. Frustration index counts *edge flips*, not unbalanced triangles. The two bad triangles might share an edge — flipping that one shared edge could fix both. So $F \leq 2$, but F could be as low as 1.

3. You need the full graph — which nodes and edges form the unbalanced triangles, and whether they share edges. Determining F exactly requires solving a minimum-cut problem on the graph (NP-hard in general, but tractable on small instances).

Takeaway

> Real signed networks are sparse and imperfect. The frustration index and the balanced-triangle fraction are the practical tools for measuring how close a network is to the balanced ideal — replacing the all-or-nothing theorem with a quantitative diagnostic.

Wrap-Up — The Four Gaps

Across Lectures 03–06, the course has exposed four kinds of gap between theory and data:

Lecture	Gap type	Theoretical ideal	Practical limitation	Strategy
L03	Computational	True tie labels (MaxSTC)	NP-hard	Polynomial proxies
L04	Computational	Optimal partition ($\max Q$)	NP-hard	GN, Louvain, Leiden
L05	Causal	Mechanism identification	Snapshots insufficient	Longitudinal data
L06	Structural	Complete balanced graph	Sparse, noisy data	Frustration index, spectral methods

Every concept in the four lectures occupies one cell of this table.

Looking Ahead: Lecture 07

How can a network be both locally clustered *and* globally navigable? Small-world networks show that a handful of random long-range shortcuts dramatically reduce path lengths — reconciling clustering and short paths without sacrificing either.

Reading

Chapter 5 of Easley & Kleinberg (2010) covers structural balance theory, weak balance, and applications. Heider (1946) and Cartwright & Harary (1956) are the original sources. Leskovec, Huttenlocher & Kleinberg (2010) is the empirical study on signed social networks.